

Benefits of Using Multiple Cameras

↳ Perfect hit detection: Adding side camera solves the Z axis problem. Instead of guessing from a flat 2d image, the system knows the exact millisecond the saber interacts with the block.

↳ No Blockage: Multiple angles eliminate blind spots. If a player crosses arms or blocks the front camera with their body, the side camera maintains the line of sight.

↳ Dynamic Scoring: Real time coordinates allow the system to calculate the exact velocity, trajectory, and angle of the swing. The game can reward players for force or accuracy of their swings.

↳ Full-Body Dodging: Multiple feeds allow the AI to generate a rough 3D bounding box of the player. This can allow dodging mechanics

Why not 1 camera?

A single camera can't accurately measure 3d depth (2-axis) which ruins the precise hit timing needed for a rhythm game. It also creates massive blind spots.

Why not 3 cameras?

Because there are IMU sensors in the sabers, they can maintain perfect tracking of the swings and solve blind spots on their own. Adding a third camera would waste processing power and complicate the setup.